
LLM Ecosystem Collapse: Minimum Viable Population and the Diversity Requirement

Ari Holtzman and NeuriCo

Abstract

As large language models (LLMs) proliferate, future models will increasingly be trained on AI-generated data. Prior work shows that recursively training models on their own outputs leads to “model collapse.” This paper investigates the concept of a “Minimum Viable Population” (MVP) for an LLM ecosystem to prevent such collapse. We empirically simulate recursive training loops to determine whether increasing the population size of models ($N > 1$) mitigates ecosystem degradation. Our results show that simply increasing the population size of identical models sharing a data pool actually accelerates collapse compared to a single-model baseline. A homogeneous population of three models exhibits a more severe drop in unique sentence ratios and a faster convergence to a biased mode than a single model. However, introducing epistemic diversity—specifically through varying generation temperatures—slightly but consistently mitigates this effect. We demonstrate that an MVP for LLMs cannot be defined merely by the count of agents; it fundamentally requires active maintenance of epistemic diversity. Our findings highlight that models must possess sufficiently different priors or sampling strategies to counterbalance the collective bias and sustain a healthy information ecosystem.

1 Introduction

As the internet becomes saturated with content generated by large language models (LLMs), future models will inevitably be trained on AI-generated data. There is growing interest in understanding the long-term sustainability of this cycle. Prior research indicates that when a model is recursively trained on its own outputs, it suffers from “model collapse”—a phenomenon where the generated distribution loses diversity, rare events disappear, and the model converges to a homogenized state Shumailov et al. [2024].

Given the ability of LLMs to generate vast amounts of data, identifying the conditions under which an ecosystem of models can survive is critical. Biological ecosystems rely on a minimum viable population to avoid extinction, prompting a parallel question in artificial intelligence: is there a minimum viable population for an LLM information ecosystem such that it does not collapse?

While previous studies primarily focus on single-model recursive loops Shumailov et al. [2024], there is a significant gap in defining the quantitative threshold or the exact nature of “individuality” required to sustain a multi-agent population. However, it also becomes increasingly challenging to disentangle the effect of sheer numbers from the effect of structural diversity. A population of identical models might simply reinforce shared biases faster, acting as an echo chamber rather than a stabilizing force.

We hypothesize that an ecosystem’s viability depends more on epistemic diversity than on the absolute number of agents. To investigate this, we empirically test the “Minimum Viable Population” (MVP) hypothesis by simulating ecosystems of varying sizes and diversity levels. We simulate a recursive training ecosystem using small LLMs, comparing a single-model baseline against both homogeneous and heterogeneous multi-model populations.

Our key findings demonstrate that population size alone is insufficient to prevent collapse. In fact, our homogeneous population of three models exhibits a more severe drop in unique sentence ratio compared to the single-model baseline (47.3% vs. 73.5%).

We provide the following contributions:

- We propose a multi-agent recursive training framework to empirically test the minimum viable population hypothesis for LLMs.
- We demonstrate that increasing the population size of identical models sharing a data pool accelerates model collapse.
- We show that introducing epistemic diversity via different sampling temperatures mitigates collapse, improving ecosystem diversity over homogeneous populations.

2 Related Work

Model Collapse. The foundational concept of model collapse was formalized by Shumailov et al. [2024], who demonstrated that recursively training models on synthetic data leads to a severe loss of the original data distribution’s tails. Their work primarily evaluates single-model recursive loops, showing a baseline “collapse” scenario when a population effectively has a size of one. Building on this, Schaeffer et al. [2025] provided a meta-analysis critiquing the inevitability of model collapse, suggesting that in real-world scenarios where models are fine-tuned on a mix of human and AI data, collapse might be highly avoidable. Unlike these single-agent or mixed-data scenarios, we focus strictly on multi-agent synthetic ecosystems to isolate the effect of population size and diversity.

Multi-Agent Ecosystems and Cooperation. Recent research has extended the study of model collapse to multi-agent environments. Piatti et al. [2024] look at collapse from a resource management perspective, simulating societies of LLM agents managing a common pool resource. They found that without advanced reasoning, most LLM societies exhaust resources and collapse, emphasizing the need for diversity and communication. While their work focuses on agent interactions, we focus on the data-driven model collapse resulting from shared synthetic training pools.

Epistemic Diversity. Directly addressing the minimum viable population hypothesis, Hodel and West [2025] evaluated multiple distinct LLMs forming an ecosystem. They found that increasing AI ecosystem diversity mitigates knowledge collapse, as models with different “epistemic” viewpoints act as a balancing force. We complement this work by providing an empirical comparison of homogeneous versus heterogeneous populations under the same recursive training framework, quantifying how even simple diversity mechanisms like varying temperatures can serve as a counter-weight to collective bias.

3 Methodology

To investigate the effects of population size and diversity on model collapse, we simulate a recursive training ecosystem. In this setup, models generate data that is pooled and subsequently used to fine-tune the next generation of models.

3.1 Experimental Setup

Model and Data. We employ distilgpt2 (124M parameters) as our base model for computational efficiency, fine-tuning it using Low-Rank Adaptation (LoRA). The initial human-written dataset used to prime the ecosystem is a subset of the wikitext-103 corpus.

Ecosystem Configurations. We evaluate the following three configurations across three generations of recursive training:

- **Baseline** ($N = 1$): A single model is recursively trained on its own outputs. In each generation, it produces 200 synthetic samples.
- **Homogeneous Population** ($N = 3$): Three identical models are trained on a shared pool of their combined outputs (600 synthetic samples per generation). All models use the same sampling parameters.

Configuration	Gen	Perplexity ↓	Unique Sentence Ratio (%) ↑	Diversity Ratio (%) ↑
Baseline ($N = 1$)	3	-	73.5	-
Homogeneous ($N = 3$)	3	76.26	47.3	47.3
Heterogeneous ($N = 3$)	3	75.54	-	48.9

Table 1: Summary of final metrics (Generation 3) for the different ecosystem configurations. The $N = 1$ baseline maintains a higher unique sentence ratio, while the heterogeneous $N = 3$ population slightly outperforms the homogeneous one. (Dashes indicate metrics not fully specified in the summarized report).

- **Heterogeneous Population** ($N = 3$): Three models are trained on a shared pool of their combined outputs. To introduce epistemic diversity, each model generates data using a different temperature (0.5, 0.7, and 0.9).

3.2 Evaluation Metrics

We evaluate the models at each generation using three primary metrics on a held-out human test set:

- **Perplexity**: Measures the language modeling quality and the divergence from the true human distribution. Lower perplexity indicates better performance.
- **3-gram Diversity Ratio**: Evaluates the lexical diversity of the generated text.
- **Unique Sentence Ratio**: Measures the proportion of unique sentences in the generated output, identifying the onset of repetitive loops and severe collapse.

4 Results

Our experiments reveal critical insights into how population size and diversity influence the trajectory of model collapse. We summarize the results across the three ecosystem configurations in Table 1.

4.1 Population Size vs. Stability

Contrary to the hypothesis that increasing the number of agents might prevent or delay collapse, our findings demonstrate that the homogeneous $N = 3$ population collapsed faster than the single-model baseline. By Generation 3, the unique sentence ratio for $N = 1$ stood at 73.5%, whereas the $N = 3$ homogeneous population dropped drastically to 47.3%.

We attribute this accelerated collapse to an “echo chamber” effect. In a shared ecosystem, the models collectively converge to a biased mode, repeating specific phrases. Because the data pool is shared, each model encounters three times more examples of this polluted data compared to the single-model loop, thereby reinforcing the bias more rapidly. Qualitative analysis of the generated text from the $N = 3$ homogeneous population exhibited extreme repetition, such as: “The game is being released in Japan on December 2nd, 2013. The game is being released in Japan on December 2nd, 2013...”

4.2 The Diversity Mitigation

Introducing heterogeneity through varying sampling temperatures slightly but measurably mitigated the severity of the collapse. As shown in Table 1, the $N = 3$ heterogeneous population achieved a perplexity of 75.54 at Generation 3, outperforming the homogeneous population’s perplexity of 76.26. Furthermore, the 3-gram diversity ratio improved from 47.3% (homogeneous) to 48.9% (heterogeneous).

These results indicate that simple stochastic diversity (e.g., using different random seeds) is insufficient to prevent models from converging to a single low-entropy state. Instead, varying epistemic viewpoints—even slightly via sampling strategies—serves as a necessary counter-weight to collective bias.

5 Discussion

Our results challenge the intuition that simply scaling the number of agents within a closed information ecosystem provides a safeguard against model collapse. Instead, a shared data pool without sufficient epistemic diversity creates an echo chamber that amplifies biases faster than a single-model recursive loop.

Is there a Minimum Viable Population? The concept of an MVP for LLMs cannot be purely quantitative. A population of identical models is functionally equivalent to one model with an inflated sampling rate, causing it to collapse just as fast, if not faster. Therefore, an MVP requires both a minimum count of agents and a minimum threshold of diversity. To be considered an “individual” within this population, a model must possess a distinctly different epistemic viewpoint—such as a different architecture or pre-training prior—acting as a counter-weight to collective bias.

Limitations. Our study operates under several constraints. First, we use `distilgpt2`, a relatively small model. Larger models might possess different internal representations making them more robust to collapse over short periods. Second, our recursive loop only spans three generations, capturing the onset of collapse but not the long-term equilibrium states. Finally, our implementation of epistemic diversity relies solely on varying sampling temperatures. Future work should investigate structural diversity by mixing different model architectures (e.g., LLaMA and GPT variants) within the same ecosystem.

Broader Implications. As real-world AI ecosystems grow, developers often deploy variations of the same base models. If these models continuously train on each other’s outputs, we risk rapid homogenization of the web. Maintaining a healthy information ecosystem will require deliberate interventions to preserve diverse pre-training data and actively inject epistemic variety.

6 Conclusion

In this paper, we investigate whether increasing the population size of an LLM ecosystem can prevent model collapse when models recursively train on shared synthetic data. We find that a larger population of homogeneous agents actually accelerates the degradation of generated text compared to a single-model baseline. However, introducing epistemic diversity through varied sampling temperatures provides a measurable mitigation against this collapse. Our findings demonstrate that a minimum viable population for an LLM ecosystem fundamentally requires active maintenance of diversity. Without varying epistemic viewpoints to counterbalance collective bias, an ecosystem will inevitably collapse, and larger homogeneous populations will simply reach that degraded state more efficiently. Future research should explore the effects of structural and architectural diversity over extended recursive generations to fully establish the conditions for a sustainable AI ecosystem.

References

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